End of Topic Test – Chapter 3: Modelling and Estimation (Test 2)

Answer all questions in the spaces provided.

1. **Fermi estimation: umbrellas on a rainy high street**

   A rainy Saturday sees 40 000 shoppers pass through a city high street throughout the day. You want to estimate how many umbrellas are being carried by shoppers on the street.

   (a) (2 points) State two assumptions you would make when estimating the number of umbrellas.

   (b) (3 points) Using your assumptions, produce a Fermi estimate for the number of umbrellas being carried.

   (c) (2 points) Explain one limitation of your estimate.

2. **Standard form and scaling**

   Light travels at approximately 3.0×10^8 m/s. A signal is sent from Earth to a Mars rover at a distance of 2.25×10^{11} m.

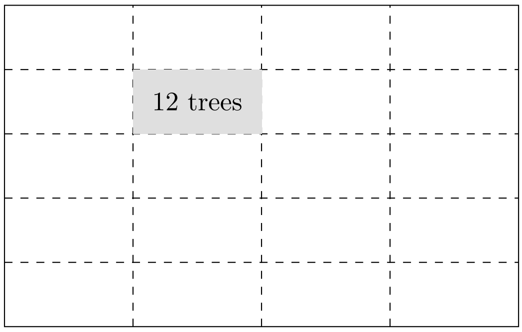
   (a) (2 points) Estimate the time (in seconds) for the signal to travel from Earth to the rover.

   (b) (3 points) Later in its orbit, Mars is three times as far from Earth. Estimate the new signal travel time, explaining your scaling method.

3. **Subdividing: estimating the number of trees in a park**

A rectangular park measures approximately 800 m by 500 m. The diagram below divides the park into a grid of equal rectangular plots. A survey of one sample plot (shaded) found 12 trees.

Park (800 m by 500 m), divided into a 4 by 5 grid

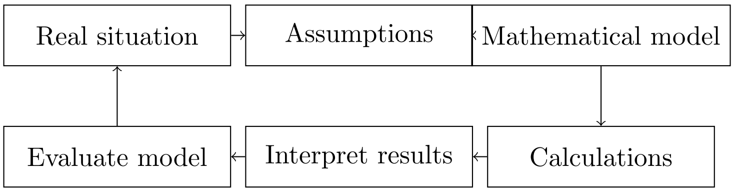


(a) (3 points) State two assumptions you need to make to estimate the total number of trees.

(b) (4 points) Use the diagram to estimate the area of one plot, and hence estimate the total number of trees in the park.

4. **Modelling cycle: estimating paper usage in a school**

   A school has 900 students. You want to estimate the total number of sheets of paper used by students during a school day.



   (a) (3 points) State three assumptions you would make when modelling paper usage.

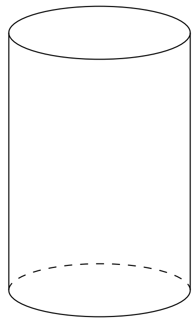
   (b) (4 points) Produce an estimate for the total number of sheets of paper used in one school day.

   (c) (2 points) Suggest one way you could improve your model.

5. **Estimating the volume of a grain silo**

A grain silo is roughly cylindrical. A diagram is shown below.

Diameter ≈ 2.4 m



Height ≈ 6 m

(a) (3 points) Estimate the volume of the grain silo in cubic metres.

(b) (2 points) Explain one limitation of modelling the silo as a perfect cylinder.

6. **Standard form: density of Jupiter**

   Jupiter has mass 1.9×10^{27} kg and volume 1.43×10^{24} m^3.

   (a) (3 points) Calculate the density of Jupiter in standard form.

   (b) (2 points) Explain why standard form is useful when working with planetary data.

7. **Assumptions: estimating the number of birds in a nature reserve**

   A nature reserve has an estimated surface area of 3.2 km^2. You want to estimate the number of birds in the reserve.

   (a) (3 points) State three assumptions you would need to make.

   (b) (4 points) Produce a Fermi estimate for the number of birds.

   (c) (2 points) Suggest one improvement to your model.